

A nocturnal increase in habituation learning performance is governed by Melatonin acting via the cooperative action of two MT1-type GPCRs.

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Abstract

Melatonin is a circadianly-regulated hormone that is produced by the pineal gland and the retina that has important roles not only in regulating sleep and circadian rhythms, but also other diverse physiological processes, including neuronal plasticity and learning. Melatonin is thought to signal primarily through two G-protein coupled receptors, MT1 (Mtnr1a) and MT2 (Mtnr1b), which are expressed in the brain and other tissues.

Introduction

Melatonin is a circadianly-regulated hormone that is produced by the pineal gland and the retina that has important roles not only in regulating sleep and circadian rhythms, but also other diverse physiological processes, including neuronal plasticity and learning. Melatonin is thought to signal primarily through two G-protein coupled receptors, MT1 (Mtnr1a) and MT2 (Mtnr1b), which are expressed in the brain and other tissues.

Results

Melatonin induces bidirectional changes in visual habituation learning

Through a high-throughput drug screen, we previously identified Melatonin as a potent modulator of habituation learning in larval zebrafish (Lamiré et al., 2023). Based on this previous dataset, we concluded

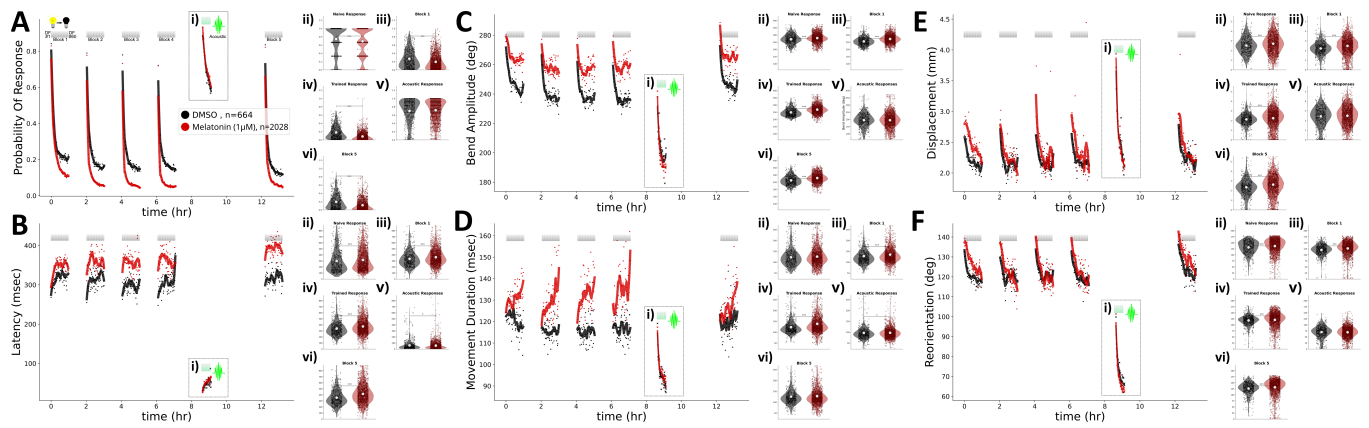


Figure 1. Melatonin has bidirectional effects on different components of habituation learning

A) Treatment with 1 μ M Melatonin (red) results in more rapid and profound decreases in the Probability of response to DF stimuli during habituation training relative to vehicle controls (0.1% DMSO black). DF stimuli are delivered at 1-minute intervals, in 4 blocks of 60 stimuli, separated by 1 hr of rest (from 0:00-7:00). 1.5 hours later a block of 30 vibration stimuli are delivered at 1-minute intervals (**i**). Each dot is the probability of response to one DF. Lines are smoothed in time with a Savitzky-Golay filter (window = 15 stimuli, order = 2). Quantification of the average.

ii)-v) Distributions responsiveness for different epochs of the experiment. Each dot is the per-fish average of the epoch. Statistical significance was calculated using the Kruskal-Wallis test followed by Dunn's test for post-hoc pairwise comparisons, * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$. **B-F)** Same as A, but for the kinematic components of the O-bend response:

B the response latency,
C the maximum tail-bend amplitude achieved during the movement,
D the duration of the movement,
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that Melatonin potently enhances habituation learning for the probability of executing an O-bend response to repeated dark-flash stimuli. Importantly, we found that Melatonin had little effect on the habituation of the magnitude of the O-bend response, which is governed by a distinct plasticity mechanisms (Randlett et al., 2019), and that Melatonin did not affect the response of the larvae to acoustic or optomotor stimuli (Lamiré et al., 2023). This demonstrates that Melatonin does not have a generalized effect on sensory processing or motor output (e.g. by simply inducing sleep Gandhi et al., 2015; Hill et al., 2026).

We have extended these analyses using a larger dataset, and confirmed that acute exposure to 1 μ M Melatonin potently enhances habituation learning for the probability of response (**Figure 1A**). Further, a significant enhancement of habituation was also observed for the response Latency, which increases by 50-100 msec over habituation training (**Figure 1B**). This effect was also visible in our original datasets (Lamiré et al., 2023), but was not previously clear and significant, likely due to the higher degree of variability in this behavioural measure. In both of these measures of behaviour, the response of the larvae to the acoustic stimuli is indistinguishable ("Acoustic Response": **Figure 1Ai,v, Bi,v**). The response to the first 3 dark-flash stimuli are not dramatically altered by the treatment ("Naive Response": **Figure 1Aii, Bii**), but the effect is pronounced during the remaining 57 flashes of the first training block ("Block 1" **Figure 1Aii, Bii**), the 180 flashes in the subsequent 3 training blocks ("Trained Response": **Figure 1Aiii, Biii**), and the re-test block 5 hours after the last training session ("Block 5": **Figure 1Avi, Bvi**). From these data we can conclude that Melatonin potentiates habituation learning for both the Probability and Latency components of behaviour.

This larger dataset now also allowed us to more carefully examine the effects of Melatonin on the behavioural components related to the magnitude of the O-bend response. Here we observed that Melatonin inhibits habituation of the Bend Amplitude (the maximum tail-angle bending achieved during the O-bend movement) (**Figure 1C**), as well as the duration of the movement (**Figure 1D**). Both of these measures normally decrease with training, resulting in movements with reduced amplitudes. As with Probability and Latency, the naive response is much less affected than the response seen during training, indicating that the primary effect is on the modulation of the response rather than their innate sensitivity. As Melatonin treated larvae showed

a weaker modulation of these components during habituation training, we conclude that Melatonin inhibits these aspects of learning, and therefore has the opposite effect on these components of behaviour compared to the Probability and Latency components.

Finally, while Melatonin inhibits habituation of the Bend Amplitude, and movement Duration, it does not have a substantial effect on the habituation of the Displacement (the distance traveled by the larvae during the O-bend movement) (**Figure 1E**) or Reorientation (the change in heading direction achieved by the larvae during the O-bend movement) (**Figure 1F**). In these two components, the naive response is slightly but significantly increased by Melatonin, consistent with a weak stimulating effect, but the responses converge during the subsequent training and testing blocks.

Collectively, these analyses of habituation learning reveal that Melatonin has a complex and bidirectional effect on different components of habituation learning, potentiating some aspects of learning while inhibiting others, and leaving others unaffected. This is consistent with a model in which Melatonin acts primarily in this context on specific neural circuits that govern distinct components of behaviour and habituation learning, rather than having a generalized effect, like modulating sleep or arousal state.

Zebrafish larvae lacking Melatonin production show habituation deficits for the Probability of response

Our pharmacological experiments have shown that exogenously applied Melatonin has a potent effect on habituation learning, but these experiments do not directly demonstrate that endogenous Melatonin is required for normal habituation learning. To address this question, we tested zebrafish mutants that are unable to produce Melatonin due to mutations in the Arylalkylamine N-acetyltransferase 1 (Aanat1) and Arylalkylamine N-acetyltransferase 2 (Aanat2) enzymes, which are expressed in the retina, and the retina and pineal gland, respectively (Gandhi et al., 2015; Hill et al., 2026). If Melatonin promotes habituation learning, we would expect that these mutants would show learning deficits. Consistent with this hypothesis, we found that double mutants (*aanat1_{-/-}; aanat2_{-/-}*) show a significant impairment in habituation learning for the Probability of response to repeated dark-flash stimuli (**Figure 2A**). Similarly to the effect of exogenous Melatonin, this effect is most pronounced during the later stages of training, and is not significant for the naive response to the first 3 dark-flash stimuli, consistent with a specific effect on learning. A significant effect was also only observed in the double mutants, indicating both Aanat1 and Aanat2 contribute to the production of Melatonin that promotes habituation learning, and that there is some degree of functional redundancy between these enzymes.

Contrary to our expectations, the Latency of the O-bend response is not significantly affected in the double mutants (**Figure 2B**). As expected, we observed no change in the kinematic parameters that are not sensitive to exogenous Melatonin (Displacement and Reorientation: **Figure 2E,F**). Similar to Latency, the kinematic components of the O-bend response that are modulated by exogenous Melatonin (Bend Amplitude, movement Duration), whose suppression is inhibited by exogenous Melatonin, are not very strongly affected in the mutants, although movement Duration did show a significantly more strongly suppressed response in double mutants (**Figure 2C-F**).

From these experiments we can conclude that endogenous Melatonin production is required for normal habituation, but this effect is only clearly apparent for the Probability of response. This is consistent with a model in which Melatonin most strongly affects this aspect of learning. However, these experiments were performed beginning in the morning, when Melatonin production is naturally low. Also, Melatonin synthesis is suppressed by light, and since our Dark Flash stimulus paradigm requires the animals to be light-adapted, little endogenous Melatonin is expected to be produced during these experiments. To try and overcome these limitations, we also performed experiments in which the larvae were raised in a shifted light cycle, such that they were tested during the circadian night when Melatonin production is naturally high. In these experiments we observed a similar pattern of results, with the most robust effect again observed for the Probability of

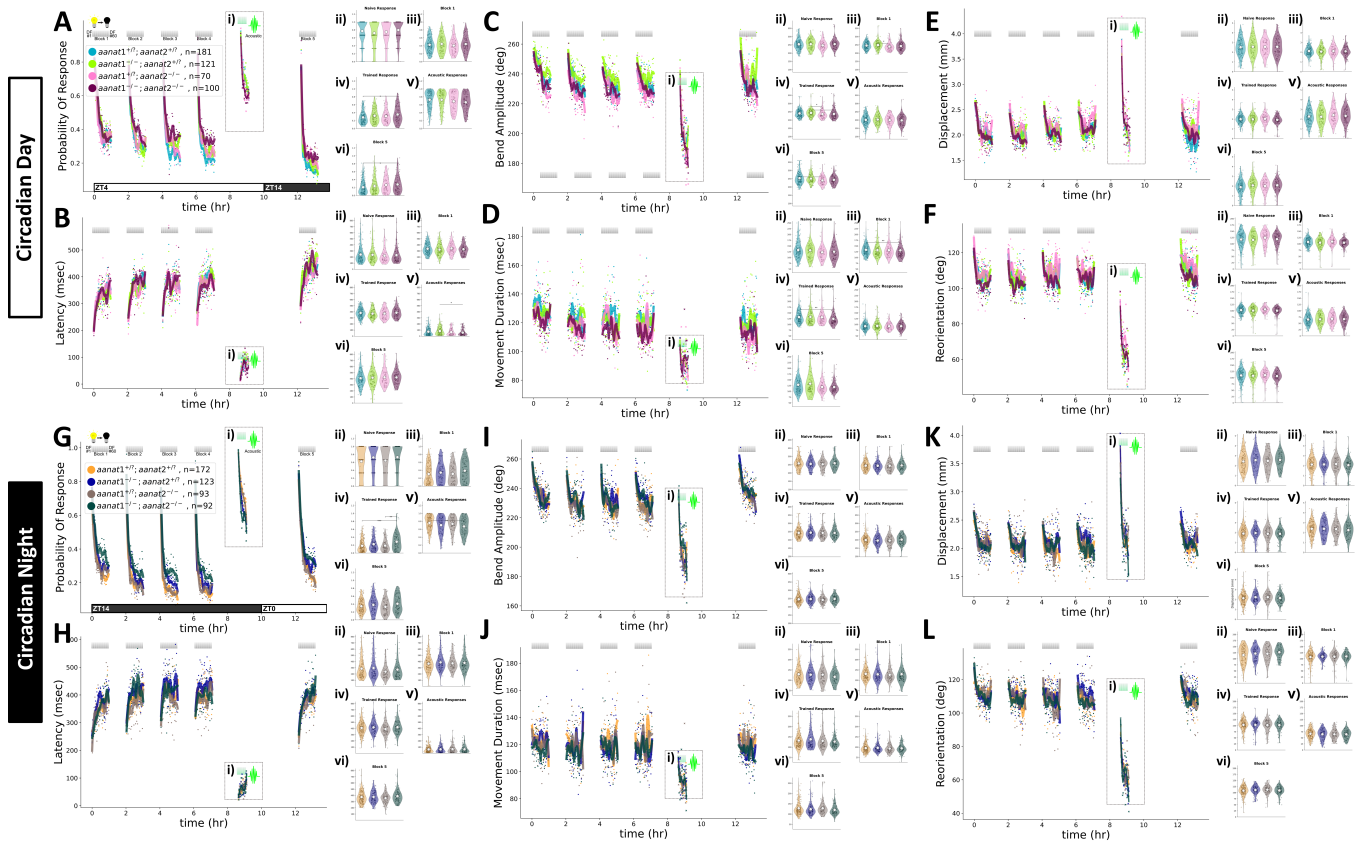


Figure 2. Zebrafish larvae lacking Melatonin production show habituation deficits for the probability of response

A) Double mutants for the melatonin-synthesizing enzymes Aanat1 and Aanat2 show a decreased modulation of the Probability of response to DF stimuli during habituation training relative to heterozygous and WT sibling controls. Testing was performed during the circadian day, beginning at approximately ZT4, where ZT0 is entrained time of lights on and ZT12 is the time of lights off. DF stimuli are delivered at 1-minute intervals, in 4 blocks of 60 stimuli, separated by 1hr of rest (from 0:00-7:00). 1.5 hours later a block of 30 vibration stimuli are delivered at 1-minute intervals (i). Each dot is the probability of response to one DF. Lines are smoothed in time with a Savitzky-Golay filter (window = 15 stimuli, order = 2).

ii)-v) Distributions responsiveness for different epochs of the experiment. Each dot is the per-fish average of the epoch. Statistical significance was calculated using the Kruskal-Wallis test followed by Dunn's test for post-hoc pairwise comparisons, * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$. **B-F)** Same as A, but for the kinematic components of the O-bend response:

B the response latency,

C the maximum tail-bend amplitude achieved during the movement,

D the duration of the movement,

E the displacement as achieved, and

F the reorientation achieved by the larvae. **G)** Double mutants for the melatonin-synthesizing enzymes Aanat1 and Aanat2 show a decreased modulation of the Probability of response to DF stimuli during habituation training relative to heterozygous and WT sibling controls. **H-L)** Same as G, but for the kinematic components of the O-bend response:

H the response latency,

I the maximum tail-bend amplitude achieved during the movement,

J the duration of the movement,

K the displacement as achieved, and

L the reorientation achieved by the larvae.

response (**Figure 2G**), and no significant effect on Latency (**Figure 2H**) or the kinematic parameters (**Figure 2I-L**). From these data we can conclude that endogenous Melatonin production is required for normal habituation learning for the Probability of response, but that its role in regulating the other learning parameters are unclear. This is likely due to the limitations of our visual assay, which are not ideal for testing the role of endogenous Melatonin as they require light-adaptation which is known to suppress Melatonin production. Therefore, for the remainder of the experiments we have focused on the effect of exogenous Melatonin on habituation learning, which is very robust and strong, and allowed us to test the genetic and circuit roles of Melatonin receptors (MtnRs).

MT1 receptors are required for the effect of Melatonin on habituation learning

Materials and Methods

Animal husbandry and care

All experiments were carried out using larval zebrafish between 5 and 6 days post-fertilization (dpf), maintained in E3 embryo medium supplemented with 0.02% HEPES (pH 7.2) at a density of approximately 1 larva per mL. Larvae were housed in 10 cm petri dishes under a 14:10 h light/dark cycle at a constant temperature of 28-29°C. Zebrafish were bred and maintained at the SickKids Zebrafish CORE facility (Toronto, Canada) and the Animalerie Zebrafish Rockefeller (AZR, SFR Santé Lyon Est, Lyon). All housing and experimental procedures were conducted in accordance with institutional and national regulations. French facilities were approved by the French Ministry of Agriculture and the Ministry of Higher Education and Research (Agreement #C693870602) and overseen by the local ethics committee (comité d'éthique en expérimentation animale de la Région Rhône-Alpes: CECCAPP). Experiments conducted in Canada were approved by the University of Toronto Animal Care Committee, in accordance with guidelines established by the Canadian Council for Animal Care (CCAC).

Generation of the *atp1a3a* mutant line

The *atp1a3a* CRISPR mutant zebrafish line was generated at the SickKids Zebrafish CORE facility (Toronto, Canada). The guide RNA (gRNA) was designed to target the *atp1a3a* gene 82 base pairs downstream of the start of exon 8. This specific region was chosen due to its close proximity to two human mutations known to cause Rapid-onset Dystonia Parkinsonism (821T>C mutation and 829G>A mutation). The resultant *atp1a3a* mutant allele has an 8 base pair deletion in exon 8 leading to a frameshift mutation with a premature stop in exon 9. For Ca²⁺ imaging experiments, *atp1a3a* mutants were crossed with *Tg(elavl3:H2B-GCaMP6f); mitfa^{-/-}*, to generate a line that express the nuclear-targeted GCaMP6f Ca²⁺ indicator pan-neuronally, in the pigment deficient *Nacre* mutants.

Genotyping

Larvae were lysed in 50 mM NaOH at 95°C for 10 min and neutralized with 1 M Tris-HCl (pH 8.0) in a 1:9 volume ratio (Tris:NaOH), to extract genomic DNA. Target region of *atp1a3a* was amplified using FIREPol® Master Mix Ready to Load 1× (Solis Biodyne) and gene-specific primers at 0.1 µM. Primer sequences used were: Forward : CCGTATTGTGGTCTGCACCG, Reverse: GCCGGTGATGATGTGGATGA. Wild-type alleles yielded amplicons of 119 bp, while mutant alleles are 111 bp, which were distinguished on an 3.5% agarose gel.

Analyses of Spontaneous Behaviour

The spontaneous behaviour of larvae was recorded using a high-speed imaging system adapted from BonZeb-based kinematic tracking (Guilbeault et al., 2021). The setup included a Sentech STC-CMB200PCL CameraLink camera (Aegis Electronic Group, Inc.) with a zoom lens (Kowa), connected via a GrabLink Full XR Frame Grabber (Euresys Inc.) for acquisition of 1088 × 1088 resolution frames at 332 Hz. Larvae were placed on a 6 mm thick transparent polycarbonate platform, illuminated from below by an 850 nm LED plate. A 1920 × 1080, 60 Hz DLP pico projector (P7, AAXA Technologies Inc.) was integrated into the system, with projected light directed onto the platform using a cold mirror (Knight Optical). Real-time video acquisition and processing were carried out in the Bonsai programming environment (Lopes et al., 2015), which interfaced with BonZeb for automated tracking (Guilbeault et al., 2021). This configuration was used to obtain free-swimming recordings of *atp1a3a* mutants, from which subsets of frames were extracted in Fiji and z-projected to generate composite images illustrating abnormal phenotypes and swim sequences. Free swimming was further analyzed at 5 dpf using the 300-well plate setup we have previously described, where larvae are recorded in 8mm x 6mm (diameter x depth) wells (Lamiré et al., 2023). Larvae were recorded over a 30-minute period in the absence of visual and acoustic stimulation. From these recordings, the mean displacement per bout (mm), mean bout duration (s), mean peak bout speed (mm/s), total displacement (mm), and bout frequency (Hz) were calculated using custom python scripts (available at github.com/owenrandlett/2025_atp1a3a).

Analyses of the Optomotor Response (OMR)

The OMR was assessed in 5dpf larvae in an acrylic arena containing 3 separate corridors allowing fish to swim in a 10.4cm x 2cm arena. Larvae were subjected to alternating vertically moving OMR gratings using a projector (AAXA LED Pico Projector) and a 45° cold mirror. OMR gratings were calculated and dynamically displayed using the “PsychoPy” package in python (Peirce, 2007). The gratings ran continually for 20 seconds epochs, alternating between directions along the long-axis of the arena, spanned by a 10 second long “no-stimulus” interval. In total, the alternating OMR stimuli ran for 10 minutes, subjecting each larva to a total of 20 repeats of the stimuli. Fish were tracked using IR LED arrays, and a camera (Chameleon 3 USB3 FLIR) running at 30 frames per second with an IR filter, and custom machine-vision scripts (available at github.com/owenrandlett/2025_atp1a3a). Lighting was projected at a screen under the arena using a projector (AAXA LED Pico Projector) and a 45° cold mirror.

Analyses of the Dark Flash Response

Larval behavior was assessed in custom 300-well acrylic plates (8 mm diameter, 6 mm depth; 300 µL water volume) based on a modified version of previously described setups. The plates were positioned beneath a 31°C water bath, which served as a heated lid to prevent condensation and maintain the wells at 29°C. Recordings were captured using a Mikrotrotron CXP-4 camera (500 Hz) connected to a Silicon Software frame grabber (Marathon ACX-QP, Basler) and illuminated with IR LEDs (TSHF5410, digikey.com). Visual stimulation was provided by a rectangular array of 155 WS2813 RGB LEDs, with dark flashes (DFs) consisting of a 1 second light-off period followed by a linear return to baseline over 20 seconds. Acoustic/vibration taps were delivered via a solenoid (ROB-10391, Sparkfun). The experimental design alternated 1 hour of stimulation with 1 hour of rest, during which a stepper motor-driven actuator (Hanpose HPV4, 500 cm) moved the camera between two plates, enabling up to 600 larvae to be tested simultaneously.

Control of the LEDs, solenoid, and camera actuator was managed by a Raspberry Pi Pico running CircuitPython and custom Python software, which also handled video acquisition and online tracking of head and tail positions at 20-30 Hz. During DF or tap delivery, a 1 second high-speed burst recording was captured and later analyzed offline using background subtraction and morphological operations. Tracking and analysis relied on open-source libraries, including OpenCV, scikit-image, NumPy, SciPy, and Numba.

MAP-Mapping Analyses of Brain-Wide Activity Patterns

Whole-brain activity mapping was performed using a MAP-mapping pipeline adapted from Randlett et al. (2015). Cohorts of 150-200 larvae produced from adult *atp1a3a*^{+/-} ; *mitfa*^{-/-} ; *Tg(elavl3:Huc-GCaMP6f)* incross were subjected to visual stimuli as following: For free swimming, larvae were grouped in one 10 cm petri dish and left swimming in a lit environment for 40 minutes in the same behavior setup as was used for the OMR stimulation (described below) to ensure identical external conditions. For OMR stimulation, larvae were grouped in one 10 cm petri dish, left to acclimatize for 10 minutes then subjected to concentric gratings that alternated vertically each 30 seconds, using the OMR behavior setup. For dark flash stimulation, larvae were grouped in one 10 cm petri dish, left to acclimatize for 20 minutes then subjected to 10 dark flashes following the dark flash protocol and using the dark flash habituation setup. Larvae were then fixed and processed for immunohistochemistry with anti-pERK (Cell Signaling Technology, #4370S) and anti-tERK primary antibodies (Cell Signaling Technology, #4696S), followed by fluorophore-conjugated secondary antibodies: Alexa FluorTM 647 for tERK (Thermo Fisher Scientific, #A21235) and Alexa FluorTM 568 for pERK (Thermo Fisher Scientific, #11036). Brains were imaged in dorsal orientation on the Leica Stellaris 5 confocal microscope equipped with a 25× water-immersion objective (HC FLUOTAR L 25x/0.95 W VISIR, CILE imaging platform, Lyon). Image stacks were acquired at a voxel resolution of $0.08 \times 0.08 \times 2 \mu\text{m}$, and two partially overlapping volumes were tiled to encompass the entire brain. The GCaMP6f channel was used as an anatomical reference for nonlinear registration of each specimen to a standardized reference brain via the Computational Morphometry Toolkit (CMTK). Registered image volumes were subsequently down-sampled to $300 \times 679 \times 80$ voxels (x, y, z) and spatially smoothed using a two-dimensional Gaussian filter. To account for inter-individual variability in staining intensity and morphology, voxel-wise pERK signals were normalized to corresponding tERK values prior to statistical comparison across experimental conditions.

Calcium imaging and head-embedded behavioural analyses

Data analysis

For the initial characterisation performed by Shahrzad, phenotypic distributions were visualized and analyzed using multiple statistical approaches.

Pie charts depicting the relative proportions of phenotypes in *atp1a3a* mutants were generated with Python packages Matplotlib, Pandas, and NumPy.

The same visualization pipeline was employed for phenotype transience assays, while histograms of tail angle distributions were constructed using these packages as well.

Statistical evaluation of swim kinematics during OMR assays was conducted with the Pingouin library.

Outliers were identified as values exceeding 1.5 times the interquartile range below the first quartile or above the third quartile and were excluded prior to subsequent analyses.

For RT-qPCR, relative expression levels of *atp1a3a* and *atp1a3b* were quantified using the $\Delta\Delta C_t$ method.

Data were analyzed with one-way ANOVAs followed by Bonferroni-corrected post hoc t-tests.

Outlier replicates were defined as those with $|z| > 3.5$ and removed prior to analysis.

A threshold of $p < 0.05$ was adopted to determine statistical significance in all tests.

Data analysis was carried out in Python using a custom-developed script.

Behavioral events were operationally defined based on cumulative tail bend angle thresholds, with deflections greater than 3 radians classified as O-bends and those exceeding 1 radian designated as C-bends.

The computational framework integrated several open-source libraries: NumPy, Pandas, Numba and SciPy for statistical procedures.

Data visualization was performed using Matplotlib and seaborn.

Statistical comparisons between distributions were conducted using the non-parametric Mann-Whitney U test as implemented in SciPy.

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